

Division Strategies — Choose Your Strategy

Stage 3 · Multiplicative relations · MA3-MR-01

NAME

PARTNER

DATE

STRATEGY 1 · SHARING AND REGROUPING



64 ÷ 4. Deal out in handfuls you already know — 10 each (that's 40 gone), then 6 each. 16 in each group.

STRATEGY 2 · KNOWN PRODUCTS

groups	× 4
1	4
10	40
5	20
16	64

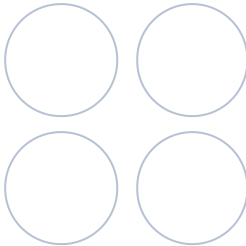
64 ÷ 4. How many 4s make 64? Write the ×10 row first, then build up. 10 + 5 + 1 = 16 groups.

A Same problem, two ways. You take one strategy, your partner takes the other — then compare.

ROUND 1 96 ÷ 4

Partner A → bubbles · Partner B → table

PARTNER A SHARING AND REGROUPING



PARTNER B KNOWN PRODUCTS

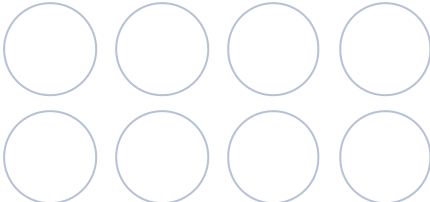
groups	× 4
1	

Which way was quicker for this one — and why?
Check it backwards: ____ × 4 = 96

ROUND 2 168 ÷ 8

now swap — A takes the table, B takes the bubbles

PARTNER B SHARING AND REGROUPING



PARTNER A KNOWN PRODUCTS

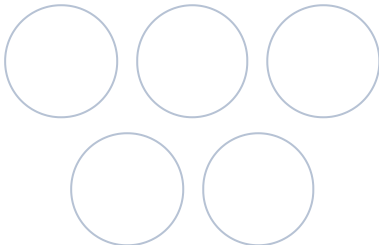
groups	× 8
1	

Which way was quicker for this one — and why?
Check it backwards: ____ × 8 = 168

ROUND 3 245 ÷ 5

swap again

PARTNER A SHARING AND REGROUPING



PARTNER B KNOWN PRODUCTS

groups	× 5
1	

Was there a quicker way than either of you used? Describe it.
Check it backwards: ____ × 5 = 245

B Now you choose. Circle the strategy you used, show it, then say why you picked it. no partner — your call

132 ÷ 6

I used: bubbles ratio table

Why that one?

375 ÷ 15

I used: bubbles ratio table

Why that one?

CHALLENGE 288 ÷ 12

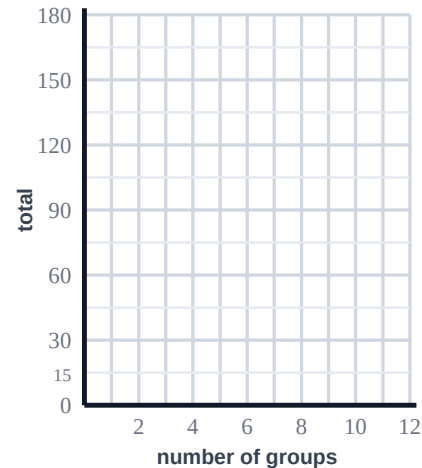
I used: bubbles ratio table

How far do bubbles get you here?

C Extension — every row is the same rate, so the rows make a line. looking ahead to Stage 4

groups	total (× 15)
1	15
2	30
5	75
10	150

This is the table you would build for 375 ÷ 15.



- 1 Plot the four rows as points — **groups** across, **total** up. Then join them with a ruler.
- 2 What do you notice about the four points?
.....
- 3 Your line passes through (0, 0). What does that point mean in this question?
.....
- 4 Every time **groups** goes up by 1, the **total** goes up by _____. Where can you see that number on the graph?
.....
- 5 Use your line, not the table: how many groups make a total of 120? _____ Now check it in the table.
- 6 Stretch. Write a rule connecting the number of groups (*g*) and the total (*t*).
.....

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ANSWER KEY — TEACHER COPY

Part A · Same problem, two ways

- $96 \div 4 = 24$ **Bubbles:** 20 each (80 gone), then 4 each. **Table:** 1|4, 10|40, 20|80, 4|16 $\rightarrow$ 24 groups. Roughly even — most will say the table.
- $168 \div 8 = 21$ **Bubbles:** 10 each (80), 10 each (160), 1 each. Eight circles is where dealing starts to drag. **Table:** 1|8, 10|80, 20|160, 1|8 $\rightarrow$ 21.
- $245 \div 5 = 49$ **The trap.** Both assigned methods are slower than halving: $245 \div 10 = 24.5$, so $245 \div 5 = 49$. A student who spots this has found strategy 3 on their own.

Round 3 is the one to listen for. Do not warn them. When a pair complains their method was silly here, that *is* the learning — put it to the room and let them state the rule: halve the divisor and the answer doubles. If nobody gets it, it is tomorrow's opening.

Part B · Now you choose

- $132 \div 6 = 22$ Either method works and that is the point — accept both, ask why. **Table:** 1|6, 10|60, 20|120, 2|12 $\rightarrow$ 22.
- $375 \div 15 = 25$ **Table, clearly.** 1|15, 10|150, 20|300 (double the row — bubbles cannot do that), 5|75 $\rightarrow$ 25. Fifteen circles is the whole argument.
- $288 \div 12 = 24$ **Challenge.** 1|12, 10|120, 20|240, 4|48 $\rightarrow$ 24. Bubbles get you started and then stall — twelve circles, and no way to double a row.

Part C · Extension — the rows make a line

- 2 They are in a straight line. (Accept “they go up evenly” — that is the same observation.)
- 3 **0 groups holds 0 altogether.** Nothing shared out yet. This is why the line starts at the corner.
- 4 **15** — the size of one group, and the unit rate row at the top of the table. On the graph it is the step up for every one step across.
- 5 **8 groups.** $120 \div 15 = 8$, and (8, 120) sits exactly on a grid intersection.
- 6 **$t = 15g$** (accept “total = groups $\times$ 15”, or $15 \times g$).

Why this belongs here. The ratio table already *is* a table of values for $t = 15g$ — students have been building linear relationships all lesson without the name. Constant rate becomes gradient, the anchor row 1 | 15 becomes the coefficient, and “through (0, 0)” is direct proportion. This is the bridge from MA3-MR-01 to MA4-RAT-C-01 and MA4-LIN-C-01.